

FAVAR vs Taylor Rule — Yield-Curve Forecasting

Last updated: 2026-05-10

Primary notebook: notebooks/favar_taylor_comparison_gsw_executed.ipynb (72 cells, fully executed)

Data: GSW zero-coupon yields + Krippner shadow short rate

What changed since our last meeting

From the discussion we had, I made four main additions to the notebook:

1. Switched the yield data to **GSW zero-coupon yields** (Gürkaynak-Sack-Wright, SVENY01–SVENY30) — these are cleaner than the raw FRED Treasury series because they're already zero-coupon and interpolated across maturities.
2. Added **Krippner's Shadow Short Rate** to handle the 2009–2015 and 2020–2021 periods where the Fed Funds rate was stuck at or near zero. During ZLB, I substitute the shadow rate for the observed FF rate.
3. Implemented the **EH Taylor Rule** as you described — the three-stage construction: forecast macro with AR(1), generate a Taylor Rule FF path with interest-rate smoothing, then use the Expectations Hypothesis to turn that path into yield forecasts.
4. Renamed the older OLS model to **Macro-OLS** so it doesn't get confused with the Taylor Rule.

How to re-run

```
python scripts/run_notebook.py notebooks/favar_taylor_comparison_gsw_executed.ipynb
```

Expected runtime: 40–60 minutes. All outputs are saved back into the notebook.

Data

Yield curve

I use GSW zero-coupon yields from the Federal Reserve (`data/feds200628.csv`), covering maturities 1Y through 30Y. The sample runs from 1985-01 to 2025-12. I process raw FRED Treasury series in parallel but the GSW notebook uses the zero-coupon data as primary.

Macro panel

35 FRED series, reduced to 25 after a 90% coverage filter. I transform flow variables (industrial production, payrolls, etc.) as log-differences annualized, and leave rates and spreads in levels. The full list includes CPI, core CPI, INDPRO, PAYEMS, Fed Funds, unemployment, T10Y2Y, VIX, and others.

For the Taylor Rule models I only use three variables: inflation (headline CPI growth), output (industrial production growth), and the Fed Funds rate. That's what Taylor (1993) originally used, and using more makes no difference — Macro-OLS actually gets worse when I add variables.

Shadow short rate

Loaded from `data/processed/ssr_monthly.csv` (Krippner 2015). Any month where the Fed Funds rate is at or below 0.25% gets the shadow rate substituted in. This matters most for 2009–2015 and 2020–2021.

Models

I run 12 models in total. The main comparison is between the **Taylor Rule variants**, **FAVAR**, and **Macro-OLS**. The rest serve as benchmarks.

Taylor Rule (EH) — professor's benchmark

Three stages:

Stage 1 — forecast inflation, output, and FF jointly using a VAR(1) expanding window.

Stage 2 — Taylor Rule with interest-rate smoothing:

$$i_t^* = r^* + \pi_t + 0.5(\pi_t - \pi^*) + 0.5y_t, \quad r^* = \pi^* = 2\%$$

$$\text{FF}_t = \rho \cdot \text{FF}_{t-1} + (1 - \rho) \cdot i_t^*$$

where ρ is estimated by OLS on the training window.

Stage 3 — Expectations Hypothesis: the yield for maturity m is just the average of the expected FF path over the next m months:

$$\hat{y}_t^{(m)} = \frac{1}{m} \sum_{j=h}^{h+m-1} \widehat{\text{FF}}_{t+j}$$

The shadow rate replaces FF at the ZLB. This is the theoretically clean construction, but it struggles — see results.

Macro-OLS

Direct OLS per tenor using the three Taylor variables:

$$\hat{y}_t^{(m)} = c_m + \rho_m \cdot y_{t-h}^{(m)} + b_1 \cdot \pi_{t-h} + b_2 \cdot \Delta \text{IP}_{t-h} + b_3 \cdot \text{FF}_{t-h}$$

No EH assumption, no iterated forecasting — just a direct h -step regression. This turns out to be competitive with the structured factor models.

Taylor Rule (EH+TP)

Same as EH but adds a constant term-premium correction per tenor. The TP is estimated as the expanding-window mean of historical yield minus the average shadow rate path. Included as a robustness check. The result is worse, not better — see §8.5.

Taylor Rule (TVR — time-varying r^*)

Same as EH except r^* at each forecast date comes from the 5-year TIPS real yield (FRED: DFII5, monthly average). Before 2003 when TIPS didn't exist, I use a constant 3% (the pre-GFC consensus). The motivation was to test whether fixing the neutral rate assumption would rehabilitate EH. It barely helps — see §8.5.

Taylor Rule (Rolling)

Same as EH but uses a 120-month rolling window instead of expanding. Includes a stability check: if the VAR companion matrix has any eigenvalue above 1, it falls back to independent AR(1) per variable.

NS-VAR

VAR(2) on first-differenced Nelson-Siegel factors $[\Delta \text{level}, \Delta \text{slope}, \Delta \text{curvature}]$. Forecasted differences are cumulated back to levels. Lambda follows a random walk in forecasting.

FAVAR

Same as NS-VAR but the state vector also includes three PCA factors from the full 25-variable macro panel:

$$[\Delta\text{level}, \Delta\text{slope}, \Delta\text{curvature}, \Delta F_1, \Delta F_2, \Delta F_3]$$

Only the NS factor forecasts are used to reconstruct yields. The macro factors are internal to the VAR.

FAVAR-Key4

Same idea but uses only 4 macro series — CPI, Fed Funds, unemployment, industrial production — and extracts 2 PCA factors from those. This was motivated by Caldeira et al. (2023). It performs better than the full FAVAR but still doesn't beat NS-VAR.

NS-RW

NS factors forecasted as a multivariate random walk ($\hat{F}_{t+h} = F_t$). No VAR dynamics at all. This is the Caldeira et al. (2023) baseline.

Diebold-Li

Fixed lambda ($\lambda = 0.0609$) Nelson-Siegel with VAR(1) dynamics on the three factors.

Yield-VAR

VAR(1) directly on the 11 raw yield levels.

Random Walk

$\hat{y}_{t+h}^{(m)} = y_t^{(m)}$. No-change benchmark.

Forecasting setup

All models use pseudo out-of-sample expanding windows. At each test date, I refit on all data up to that point and forecast $h = 1, 6$, and 12 months ahead. The test window is December 2015 to December 2025 (119 months).

For VAR-based models I iterate the forecast forward. For Macro-OLS and the Taylor Rule variants I use direct h -step OLS with regressors lagged by h periods — no iteration.

Results

Average RMSE across 11 tenors

Model	h=1	h=6	h=12	vs RW (h=1)	vs RW (h=12)
Yield-VAR	0.1937	0.6726	1.1728	0.94	0.93
NS-VAR	0.2049	0.7782	1.3461	0.99	1.07
Macro-OLS	0.2053	0.7650	1.2820	1.00	1.02
Random Walk	0.2065	0.7478	1.2590	1.00	1.00
NS-RW	0.2105	0.7491	1.2596	1.02	1.00
FAVAR-Key4	0.2092	0.7781	1.3473	1.01	1.07
Diebold-Li	0.2222	0.6757	1.1415	1.08	0.91
FAVAR	0.2674	0.8803	1.4183	1.30	1.13
Taylor Rule (Rolling)	1.0439	1.2907	1.7022	5.05	1.35
Taylor Rule (EH)	1.3857	1.6821	2.1612	6.71	1.72
Taylor Rule (TVR)	1.3775	1.6970	2.2066	6.67	1.75
Taylor Rule (EH+TP)	2.8836	3.1801	3.5837	13.96	2.85

Yield-VAR is the best model at h=1 and h=6. Diebold-Li wins at h=12. Macro-OLS and NS-VAR are essentially tied with the Random Walk at h=1 — they don't hurt but they don't help much either at short horizons.

RMSE relative to Random Walk

Model	h=1	h=6	h=12
Yield-VAR	0.94	0.90	0.93
NS-VAR	0.99	1.04	1.07
Macro-OLS	1.00	1.02	1.02
Random Walk	1.00	1.00	1.00
NS-RW	1.02	1.00	1.00
FAVAR-Key4	1.01	1.04	1.07

Model	h=1	h=6	h=12
Diebold-Li	1.08	0.90	0.91
FAVAR	1.30	1.18	1.13
Taylor Rule (Rolling)	5.05	1.73	1.35
Taylor Rule (EH)	6.71	2.25	1.72
Taylor Rule (TVR)	6.67	2.27	1.75
Taylor Rule (EH+TP)	13.96	4.25	2.85

EH Taylor Rule variants — four robustness checks

Variant	h=1 RMSE	h=6 RMSE	h=12 RMSE	vs EH raw	Takeaway
Taylor Rule (EH)	1.3857	1.6821	2.1612	baseline	Fixed $r^*=2\%$, expanding window
Taylor Rule (TVR)	1.3775	1.6970	2.2066	-0.6%	Time-varying r^* : barely changes anything
Taylor Rule (Rolling)	1.0439	1.2907	1.7022	-25%	Adaptive window helps, confirms regime instability
Taylor Rule (EH+TP)	2.8836	3.1801	3.5837	+108%	Constant TP from training era makes it much worse

I ran these four variants to understand where exactly EH breaks down. The answer is consistent across all of them: the problem isn't the neutral rate, and it isn't a missing term premium — it's the EH assumption itself. Averaging expected short rates gives you a yield forecast that mechanically ignores the term premium, which is 100–300 bps on long yields. You can't fix that with a parameter adjustment.

The TVR result is the clearest evidence. Even with r^* at 0.19% in 2015 and 0.78% in 2018 — values consistent with Holston-Laubach-Williams estimates — the model barely improves (0.6%). Adding a constant term premium from the training era makes it worse because the historical 30Y TP was around 5.98% but actual 30Y yields during the test period were 2–4%.

Sub-period breakdown (h=1, RMSE relative to RW)

Model	Pre-COVID (2016–2019)	ZLB/COVID (2020–2021)	Hiking Cycle (2022–2025)
Random Walk	1.000	1.000	1.000
Yield-VAR	~1.00	~0.95	0.887
Macro-OLS	0.964	~1.00	0.900
NS-VAR	~0.98	0.937	~0.97
FAVAR-Key4	~1.02	~0.97	~1.00
Taylor Rule (EH)	6.35	11.97	3.89

A few things stand out here:

- The EH Taylor Rule is worst during ZLB/COVID ($11.97\times$ RW). This makes sense — that's when the gap between $r^*=2\%$ and the actual rate is largest.
- Yield-VAR's advantage is almost entirely in the Hiking Cycle ($0.887\times$ RW). The 2022–2025 rapid tightening was highly persistent, which is exactly the kind of environment where a simple autoregression on yield levels does well.
- Macro-OLS is competitive in both the pre-COVID and hiking cycle periods, when the Fed was actively responding to inflation and output signals.

FAVAR-4F robustness

Horizon	FAVAR-3F	FAVAR-4F
h=1	0.2674	0.2751
h=6	0.8803	0.9381
h=12	1.4183	1.4702

Adding a fourth macro PCA factor makes FAVAR worse at every horizon. More macro information doesn't help.

Statistical tests

I use pairwise Diebold-Mariano tests (Harvey-Leybourne-Newbold 1997) with Newey-West HAC variance. A positive DM stat means model B has lower MSE.

Comparison	h=1 avg DM	Tenors significant (p<0.10)
NS-VAR vs EH Taylor Rule	+8.55	11/11
FAVAR vs EH Taylor Rule	+7.84	10/11
NS-VAR vs Random Walk	-1.92	0/11
FAVAR vs NS-VAR	-1.77	0/11
NS-RW vs NS-VAR	+1.56	5/11
FAVAR-Key4 vs FAVAR	+1.12	0/11
NS-VAR vs Macro-OLS	~0	0/11

NS-VAR and FAVAR both beat the EH Taylor Rule by a wide margin at $h=1$ — significant across all tenors. But neither beats the Random Walk significantly. NS-RW beats NS-VAR at 5 tenors, which is consistent with the Caldeira et al. (2023) finding that imposing VAR dynamics on NS factors adds noise rather than information.

One thing still to do: formal DM tests for Diebold-Li and Yield-VAR vs RW at $h=12$. Because $h=12$ predictions overlap, the standard DM may be mis-sized — Clark-McCracken (2001) is the right test here.

Conclusions

1. **At $h=1$, yield-level persistence matters more than macro information.** Yield-VAR (RMSE 0.19, 6% below RW) is the clear winner. NS-VAR, Macro-OLS, and the Random Walk are all within 0.01 of each other.
2. **At $h=12$, Diebold-Li wins** (RMSE 1.14, 9% below RW). Some macro signal comes through at longer horizons.
3. **EH Taylor Rule fails throughout.** At $h=1$ it's $6.7\times$ worse than the Random Walk. The problem is structural — the EH assumption gives zero weight to the term premium, which matters especially for long yields.
4. **Four robustness checks all point to the same conclusion.** Time-varying r^* (TVR) barely improves things. A constant term-premium correction makes it worse. A rolling window helps 25% but the model still fails. The EH framework itself is the issue, not any particular parameter.
5. **FAVAR does not improve on NS-VAR.** Adding the full 25-variable macro panel to the VAR increases RMSE from 0.20 to 0.27 at $h=1$. FAVAR-Key4 (4 focused variables) narrows that gap but still loses.
6. **Macro-OLS matches NS-VAR at $h=1$ with just three variables.** This is the parsimony result: three strict Taylor variables in a direct OLS outperforms a 6-variable factor VAR across most settings.
7. **Regime matters a lot.** The EH failure is concentrated in 2020–2021 when the ZLB was binding ($11.97\times$ RW). Yield-VAR's advantage is concentrated in 2022–2025 during the hiking cycle ($0.887\times$ RW).

Open questions

Why does Macro-OLS beat FAVAR?

Both use macro information. A few possible explanations:

- FAVAR has more parameters and is less stable out-of-sample despite the larger training window
- PCA compresses 25 variables into 3 factors, which may dilute the most relevant signals (especially Fed Funds)
- Macro-OLS uses a direct h -step regression while FAVAR iterates — compounding errors matter at $h=12$

The cleanest test would be FAVAR with direct h -step OLS instead of iterated VAR, to separate the variable selection effect from the forecasting method.

Does time-varying r^* fix the EH model? — Answered

No. Using TIPS-based r^* (0.19% in 2015, 0.78% in 2018) reduces $h=1$ RMSE from 1.3857 to 1.3775 — a 0.6% improvement. The EH no-term-premium assumption is the dominant failure, not r^* calibration.

Statistical tests at $h=12$

Diebold-Li ($0.91\times$ RW) and Yield-VAR ($0.93\times$ RW) both beat the Random Walk at $h=12$. These haven't been formally tested yet. Given overlapping 12-month forecasts, the Clark-McCracken (2001) test is needed before claiming significance.

No-arbitrage model

The EH+TP diagnostic rules out a constant TP correction. To properly rehabilitate EH you'd need time-varying risk prices — an Ang-Piazzesi (2003) type model. That's a different scope of work. For the thesis, I'd frame the EH failure and the diagnostic results as documenting *why* EH breaks down, which motivates that literature without requiring implementation.

File map

File	Description
data/feds200628.csv	GSW zero-coupon yields SVENY01–SVENY30

File	Description
data/yields_raw.csv	11-tenor FRED Treasury yields
data/macro_raw.csv	35 FRED macro series
data/processed/ssr_monthly.csv	Krippner Shadow Short Rate (monthly)
data/rstar_monthly.csv	TIPS-based r^* series (DFII5 monthly avg, 2003–present; 3.0% pre-2003)
notebooks/favar_taylor_comparison_gsw_executed.ipynb	Primary results notebook (72 cells, fully executed)
notebooks/favar_taylor_comparison_executed.ipynb	Older FRED-yield notebook (61 cells)
notebooks/fetch_data.ipynb	Downloads data from FRED
docs/favar_taylor_pipeline_professor_review.md	This document